Tutorial VI - Minimizing Split Orders in E-Commerce

Applied Optimization with Julia



Introduction

> **Optional: A Zed Theme for the Tutorials**
> In these tutorial files, the comments are the instructions, yet most Zed themes grey them out. The theme Paper & Ink does the opposite: it gives the comments the strongest contrast and the code calmer colors. Run the next cell once to install it. Then quit Zed completely (macOS: `Cmd+Q`), start it again, open the command palette (`Ctrl+Shift+P`, macOS `Cmd+Shift+P`), run **theme selector: toggle** and choose **Paper & Ink Light** or **Paper & Ink Dark**.

```
using Downloads
themes = Sys.iswindows() ? joinpath(ENV["APPDATA"], "Zed", "themes") :
expanduser("~/.config/zed/themes")
mkpath(themes)
Downloads.download("https://raw.githubusercontent.com/beyondsimulations/paper-and-ink-theme/main/themes/paper-and-ink.json", joinpath(themes, "paper-and-ink.json"))
```

We are consulting for FitGear, a rapidly growing e-commerce company specializing in fitness equipment and athletic wear. FitGear is facing a challenge: customer complaints about split orders have increased by 40% in the last quarter. When a customer orders multiple items, they sometimes receive multiple packages from different warehouses, leading to higher shipping costs and frustrated customers.

This tutorial explores how to optimize SKU allocation across warehouses to minimize split orders using mathematical optimization.

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1. Coappearance Matrix

First, we need to understand which products are frequently purchased together. We'll use a coappearance matrix. It shows how frequently items appear together in customer orders. For each pair of items i and j, the matrix element $q_{i,j}$ represents the number of orders containing both items i and j. Let's start with a small sample of FitGear's historical order data and determine the coappearance matrix of the transactional data set **by hand**.

$t_{m,i}$	A	B	C	D
1	1	1	1	0
2	1	1	1	0

$t_{m,i}$	A	B	C	D
3	1	1	0	0
4	1	0	0	1
5	1	0	0	1
6	1	0	0	1
7	1	0	0	1
8	0	0	1	1

What is the resulting **coappearance matrix** from the transactional data set? Please provide it in the following cell as matrix by changing the `0`s to the correct values.

```
Q = [
    0 0 0 0;
    0 0 0 0;
    0 0 0 0;
    0 0 0 0;
]
```

---

2. Maximizing Coappearances

FitGear currently operates two warehouses - one in Hamburg and one in Berlin. Each warehouse has limited storage capacity:

- Hamburg can store 3 different product types (SKUs)
- Berlin can also store 3 different product types

Based on the coappearance matrix you have determined in the previous assignment, use the model from the lecture to maximize the coappearances **by hand**. Note one difference to the lecture: assume that each SKU may only be stored in **one** warehouse, i.e. the allocation constraint holds with equality. What is the resulting objective function value of the assignment? Please provide it in the following cell.

```
objective_value_by_hand =
```

---

3. Counting Split Orders

The objective value alone does not tell us the number of split orders. Calculate the **number of split orders by hand** from your previous solution.

```
number_of_split_orders =
```

> **Multiple Optimal Solutions**
> Several different allocations reach the same optimal objective value here. Depending on which one you found, your number of split orders can differ! If possible, compare your count with a neighbor: did you get the same objective value but a different number of split orders? Keep this in mind for the bonus question.

> **Bonus Question**
> Is the solution of the Quadratic Multiple Knapsack Problem (QMKP) the optimal solution for the underlying split order minimization problem? Try to find another allocation with fewer (or the same number of) split orders.

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4. Implementing the model

As FitGear continues to grow, the operations team needs a solution that can handle their full product catalog. They now have:

- 12 popular fitness products
- 40 recent customer orders
- Two warehouses with different capacities:
  - Hamburg: 7 SKUs
  - Berlin: 6 SKUs

Thus, we'll now implement a scalable solution using Julia and JuMP that can handle larger datasets.

```
skus = [
    "Running Shoes",
    "Athletic Socks",
    "Water Bottle",
    "Yoga Mat",
    "Resistance Bands",
    "Sports Bag",
    "Protein Powder",
    "Exercise Shorts",
    "Training Shirt",
    "Fitness Tracker",
    "Foam Roller",
    "Weight Gloves"
]

warehouses = ["Hamburg", "Berlin"]

capacity = [7, 6]

T = [
    # First set of transactions (1-20)
    1 1 0 0 0 0 0 0 0 0 0 0;  # shoes + socks
```

```
    1 1 0 0 0 1 0 0 0 0 0 0;  # shoes + socks + sports bag
    0 0 0 1 1 0 0 0 0 0 1 0;  # yoga mat + bands + foam roller
    0 0 1 1 0 0 0 0 0 0 0 0;  # water bottle + yoga mat
    0 0 0 0 0 0 1 0 0 0 0 0;  # just protein powder
    0 0 0 0 0 0 1 0 0 0 0 1;  # protein powder + gloves
    0 0 0 0 0 0 0 1 1 0 0 0;  # shorts + shirt combo
    0 0 0 0 1 0 0 1 1 0 0 0;  # workout outfit + bands
    0 0 0 0 0 0 0 0 0 1 0 0;  # just fitness tracker
    1 1 1 0 0 1 0 1 1 0 0 0;  # full running gear set
    0 0 0 1 1 0 0 0 0 0 1 1;  # home gym basics
    0 0 0 0 0 0 1 1 1 0 0 1;  # gym starter pack
    0 0 1 0 1 0 0 0 0 0 0 0;  # water bottle + bands
    1 1 0 0 0 0 0 1 1 0 0 0;  # running outfit complete
    0 0 0 1 1 0 0 0 0 0 1 0;  # yoga equipment set
    0 0 0 0 0 0 1 0 0 0 0 1;  # protein powder + gloves
    0 0 1 0 0 0 1 0 0 0 0 0;  # water bottle + protein
    1 0 0 0 0 1 0 0 0 1 0 0;  # shoes + bag + tracker
    0 0 0 0 1 0 0 1 1 0 0 0;  # workout outfit + bands
    0 0 0 1 0 0 0 0 0 0 1 0;  # yoga mat + foam roller
    # Second set of transactions (21-40)
    1 1 0 0 0 0 0 1 1 0 0 0;  # shoes + socks + workout clothes
    1 1 0 0 0 1 0 0 0 1 0 0;  # shoes + socks + bag + tracker
    0 0 0 1 1 0 0 0 0 0 1 1;  # yoga mat + bands + foam roller + gloves
    0 0 1 1 1 0 0 0 0 0 0 0;  # water bottle + yoga mat + bands
    0 0 1 0 0 0 1 0 0 0 0 1;  # protein powder + water bottle + gloves
    0 0 0 0 1 0 1 0 0 0 0 1;  # protein powder + bands + gloves
    0 0 0 0 0 0 0 1 1 1 0 0;  # shorts + shirt + tracker
    1 0 0 0 1 0 0 1 1 0 0 0;  # shoes + workout outfit + bands
    0 0 1 0 0 0 0 0 0 1 0 0;  # water bottle + tracker
    1 1 1 0 0 1 0 1 1 1 0 0;  # deluxe running gear set
    0 0 0 1 1 0 1 0 0 0 1 1;  # advanced home gym set
    0 0 1 0 0 0 1 1 1 0 0 1;  # gym starter pack with water bottle
    0 0 1 0 1 0 0 0 0 1 0 0;  # water bottle + bands + tracker
    1 1 0 0 0 1 0 1 1 0 0 0;  # running outfit with bag
    0 0 0 1 1 0 0 0 0 0 1 1;  # yoga equipment set with gloves
    0 0 1 0 0 0 1 0 0 0 1 1;  # protein set with foam roller
    0 0 1 0 0 1 1 0 0 0 0 0;  # water bottle + protein + bag
    1 0 0 0 0 1 0 0 0 1 1 0;  # shoes + bag + tracker + foam roller
    0 0 0 0 1 0 0 1 1 1 0 0;  # workout outfit + bands + tracker
    0 0 0 1 0 0 0 0 0 1 1 0   # yoga mat + tracker + foam roller
]
```

Implement the model in Julia and solve it for the given data set.

> **Each SKU can only be allocated to one warehouse!**
> The retailer has another requirement: each SKU can only be allocated to one warehouse. Make sure to include this requirement in your model.

Use the Juniper solver

If you don't have Juniper, Ipopt and/or HiGHS installed, add the solvers via `Pkg.add("Juniper")`, `Pkg.add("Ipopt")` and `Pkg.add("HiGHS")`. Juniper is a solver for nonlinear problems that can be used in combination with Ipopt and HiGHS to solve mixed-integer quadratic problems.

> **Note**
> If you want to use the solver SCIP, you can also do this by adding JuMP and SCIP and then change the solver in the model definition to `warehouse_model = Model(SCIP.Optimizer)`. Note that this does not work automatically in Windows, as you will have to install the SCIP binaries manually. Hence, I recommend sticking with Juniper. On Mac and Linux, SCIP should work out of the box.

First, we start by defining the model.

```
# Definition of the warehouse model
using JuMP, Ipopt, HiGHS, Juniper
ipopt = optimizer_with_attributes(Ipopt.Optimizer, "print_level" => 0)
highs = optimizer_with_attributes(HiGHS.Optimizer, "output_flag" => false)
warehouse_model = Model(
    optimizer_with_attributes(
        Juniper.Optimizer,
        "nl_solver" => ipopt,
        "mip_solver" => highs,
    ),
)
```

```
A JuMP Model
├ solver: Juniper
├ objective_sense: FEASIBILITY_SENSE
├ num_variables: 0
├ num_constraints: 0
└ Names registered in the model: none
```

Compute the coappearance matrix

Next, compute the coappearance matrix based on the transactional data provided in `T` and call it `Q`.

```
# YOUR CODE BELOW
```

```
# Assert whether the coappearance matrix is correct.
@assert Q == Q' "The coappearance matrix is not symmetric. Have you transposed
the transactional data set?"
@assert Q[1,1] == 11 "The coappearance matrix is not correct. Have you
multiplied the transposed transactional data set with itself?"
println("Great! The coappearance matrix is correct.")
```

Define the decision variable

Now, define the decision variable for the SKU allocation. Please name the variable `X`.

```
# YOUR CODE BELOW
```

```
# Assert whether the decision variable is correct.
@assert typeof(X) <: AbstractArray{VariableRef} "The decision variable X should be an array of JuMP variables"
@assert all(is_binary.(X)) "The decision variable X should be defined as binary"
@assert size(X) == (length(skus), length(warehouses)) "The decision variable X should have dimensions [skus × warehouses]"
println("Great! The decision variable is correctly defined as a binary variable with proper dimensions.")
```

Define the objective function

Then, define the objective function to maximize the coappearance.

> **Mixing names and indices**
> If you followed the pattern from the lecture, `X` is indexed by the SKU and warehouse *names*, while `Q` uses *integer* indices. In the objective function, you can combine both by looking up the names, e.g. `X[skus[i], warehouses[k]]` together with `Q[i,j]`.

```
# YOUR CODE BELOW
```

```
# Assert whether the objective function is correct.
@assert typeof(objective_function(warehouse_model)) <: QuadExpr "The objective function should be a quadratic expression of JuMP variables"
println("Great! The objective function is correctly defined.")
```

Define the constraints

To ensure that each SKU is allocated to one warehouse, add the first constraint.

```
# YOUR CODE BELOW
```

```
# Assert whether the single allocation constraint is correct.
@assert num_constraints(warehouse_model, AffExpr, MOI.EqualTo{Float64}) == length(skus) "The single allocation constraint should have one constraint for each SKU. Did you use >= as in the lecture instead of ==? Here, each SKU must be allocated to exactly one warehouse."
println("Great! The single allocation constraint is correctly defined.")
```

To ensure that the capacity of each warehouse is not exceeded, add the second constraint.

```
# YOUR CODE BELOW


# Assert whether the capacity constraint is correct.
@assert num_constraints(warehouse_model, AffExpr, MOI.LessThan{Float64}) ==
length(warehouses) "The capacity constraint should have one constraint for each
warehouse"
println("Great! The capacity constraint is correctly defined.")
```

Solve the model

Finally, solve the model with a solve statement.

> ⚠ **Juniper only guarantees local optimality**
> Juniper is a *local* solver for this kind of problem, so the status `LOCALLY_SOLVED` does not guarantee the best possible solution. The optimal objective value is 141. If your model reports a lower value, your code can still be correct: re-run the solve statement or try the SCIP solver, which proves global optimality.

```
# YOUR CODE BELOW


# Assert whether the model is solved correctly.
@assert termination_status(warehouse_model) == MOI.OPTIMAL ||
termination_status(warehouse_model) == MOI.LOCALLY_SOLVED "The model should
either be solved with an optimal solution or with a locally optimal solution"
@assert isapprox(objective_value(warehouse_model), 141) "The objective value
should approximately be 141. If it is lower and the status is LOCALLY_SOLVED,
see the warning above. Otherwise, have you correctly implemented the objective
function?"
println("Great! The model is correctly solved.")
```

Print the results

The following code prints the objective value, the SKU allocation and the warehouse capacities based on your optimal solution.

```
println("Objective value: ", objective_value(warehouse_model))
println()
println("SKU Allocation:")
for i in skus
    for k in warehouses
        if value(X[i,k]) > 0.5
            println("SKU: ", i, " Allocation: ", k)
        end
    end
```

```
end
println()
println("Warehouse Capacities:")
for k in eachindex(warehouses)
    println("Warehouse: ", warehouses[k], " Capacity: ", capacity[k], " Used:
", sum(value(X[i,warehouses[k]]) for i in skus))
end
```

Count the number of split orders

The objective value alone still does not tell us the number of split orders. The following code calculates the number of split orders from the SKU allocation for the optimal solution. To judge how good this is, it also computes the average number of split orders over 100 random allocations. Note that the random allocations are still *feasible*: each SKU is stored in exactly one warehouse and no capacity is exceeded — otherwise, the comparison would not be fair.

```
# Binary matrix indicating whether a SKU is allocated to a warehouse.
X_values = [value(X[i,k]) > 0.5 for i in skus, k in warehouses]

# Function to count the number of split and regular orders.
function count_split_orders(X_input, transactional_data)
    # Initialize the counters for split and regular orders.
    split_orders = 0
    regular_orders = 0

    # Iterate over each transaction in the transactional data set.
    for t in 1:size(transactional_data,1)

        # Check whether any warehouse can fulfill the transaction alone.
        if any(all(X_input[:,k] .>= transactional_data[t,:]) for k in
axes(X_input,2))
            regular_orders += 1

        # If no warehouse can fulfill the transaction alone, it is a split.
        else
            split_orders += 1
        end
    end
    return split_orders, regular_orders
end

# Count the number of split and regular orders for the optimal solution.
split_orders, regular_orders = count_split_orders(X_values, T)

# Count the split and regular orders for 100 random feasible allocations.
using Random
```

```
split_orders_random = []
regular_orders_random = []
for trial in 1:100
    # Shuffle the SKUs, then fill up Hamburg first and Berlin afterwards.
    shuffled_skus = shuffle(1:length(skus))
    X_random = zeros(Bool, length(skus), length(warehouses))
    for (position, i) in enumerate(shuffled_skus)
        if position <= capacity[1]
            X_random[i,1] = true
        else
            X_random[i,2] = true
        end
    end
    split_random_trial, regular_random_trial = count_split_orders(X_random, T)
    push!(split_orders_random, split_random_trial)
    push!(regular_orders_random, regular_random_trial)
end
split_orders_random = sum(split_orders_random)/100
regular_orders_random = sum(regular_orders_random)/100

# Print the number of split and regular orders.
println("Number of split orders (optimal): ", split_orders)
println("Number of regular orders (optimal): ", regular_orders)
println()

# Print the number of split and regular orders for the random allocations.
println("Number of split orders (random): ", split_orders_random)
println("Number of regular orders (random): ", regular_orders_random)
```

> **Split orders and split parcels**
> With two warehouses, each split order causes exactly one extra parcel. With more warehouses, a single split order could require several extra parcels, so the two numbers would no longer coincide.

---

5. Analyzing the Results

FitGear's operations team needs to understand the business implications of this optimization:

- Which product categories tend to cluster together in the same warehouse? For example, do workout clothes tend to be stored together?
- How many fewer split shipments would FitGear have compared to their current random allocation?
- How might this new allocation affect warehouse operations and picking efficiency?

Your answer goes here, thinking from the business perspective. A few sentences are fully sufficient!

```
# YOUR ANSWER BELOW
#=


=#
```

Solutions

You will likely find solutions to most exercises online. However, I strongly encourage you to work on these exercises independently without searching explicitly for the exact answers to the exercises. Understanding someone else's solution is very different from developing your own. Use the lecture notes and try to solve the exercises on your own. This approach will significantly enhance your learning and problem-solving skills.

Remember, the goal is not just to complete the exercises, but to understand the concepts and improve your programming abilities. If you encounter difficulties, review the lecture materials, experiment with different approaches, and don't hesitate to ask for clarification during class discussions.

Bibliography